DO NOW — RECALL

Lesson 064 · Thinking Concurrently: Independent Work

Name: _____ Class: _____ Date: _____

1. What is a process dependency?

2. What is throughput?

3. What is a race condition?

4. A rising-edge D-type flip-flop starts with Q=0. At a rising edge D=1, so Q becomes 1. Immediately afterwards D becomes 0. Explain why Q stays 1 through the following falling edge.

DO NOW — CHECK / OBJECTIVES

Correct and improve

Check page 1 in GREEN PEN. Write corrections in full.

1. Model answer

A task requires another task's result.

2. Model answer

Amount of completed work per unit time.

3. Model answer

Result depends on unsafe timing/order of shared operations.

4. Model answer

Only a rising edge samples D. Changing D and reaching a falling edge do not update Q, so it holds 1 until the next rising edge samples 0.

By the end, you can…

• Find tasks that do not depend on each other's intermediate results. Independent work can overlap.

• Distinguish concurrency (progress interleaved) from parallelism (truly simultaneous hardware execution).

• Evaluate scheduling overhead and shared-resource/race risks, and explain why dependencies limit possible speedup.

Vocabulary — know these terms

concurrency — Multiple tasks make progress in overlapping period.

parallelism — Actual simultaneous execution.

critical path — Longest dependency chain.

race condition — Incorrect timing-dependent shared state.

NEW LEARNING

Read, annotate and explain

Identify

Find tasks that do not depend on each other's intermediate results. Independent work can overlap.

Model

Distinguish concurrency (progress interleaved) from parallelism (truly simultaneous hardware execution).

Evaluate

Evaluate scheduling overhead and shared-resource/race risks, and explain why dependencies limit possible speedup.

Explain one step in your own words, using the correct technical terms.

I DO — WORKED EXAMPLE

Follow the worked method

Calculate minimum ideal time for independent image operations 6s and 4s on separate cores.

Explain the decision

Only independent tasks can safely overlap without waiting for each other.

WE DO — GUIDED PRACTICE

Try → Discuss → Improve

YOU DO — INDEPENDENT PRACTICE

Questions · Part 1

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 1

Independent jobs take 7s and 4s. Calculate sequential and ideal parallel completion times. [4]

Question 2

Give two potentially concurrent stages of a school app and two stages that must be sequential. [6]

YOU DO — INDEPENDENT PRACTICE

Questions · Part 2

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 3

Two threads increment shared counter via read, add, write and occasionally lose updates. Explain race and one prevention. [6]

EXIT TICKET / REFLECTION

Check your understanding

Complete the actual lesson Exit Ticket before checking the answer.

Doubling the number of CPU cores always halves task completion time.

How confident are you? Circle one: Confident / Getting there / Need help

Explain your choice, identifying one concept you can now use independently.

GREEN PEN — Correct and explain

Choose one mistake from your independent work. Write the correct method and explain precisely why your original answer was wrong.

Next step: revisit the model or a relevant revision-guide section if needed.